

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Summer Regular Examination – 2023

Course: B. Tech. Branch :CSE/Computer Engineering Semester : VI

Subject Code & Name: BTCOC603 Machine Learning

Max Marks: 60 Date:17/07/2023 Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Define machine learning. Explain its common areas of applications.	(Remember)	6
B) Describe working of decision tree with suitable example.	(Apply)	6
C) State the purpose of confusion metrics. Elaborate various terms associated with it.	(Understand)	6
Q.2 Solve Any Two of the following.		12
A) State Bayestheorem. Demonstrateit with suitable example.	(Apply)	6
B) With respect to SVM explain the terms: 1. Hyperplane 2. Maximum Margin Hyperplane (MMH) 3. Support Vector.	(Understand)	6
C) Differentiate between linear and non-linear SVM.	(Analyze)	6
Q. 3 Solve Any Two of the following.		12
A) Compare between Biological Neural Network (BNN) and Artificial Neural Network (ANN).	(Analyze)	6
B) Explain reinforcement learning in Artificial Neural Network (ANN).	(Understand)	6
C) Describe feedforward and feedback neural network with proper diagram.	(Understand)	6
Q.4 Solve Any Two of the following.		12
A) Explain concept of VC dimensions with example.	(Remember)	6
B) Discuss Ensembled Learning. State applications of ensembled learning.	(Understand)	6
C) Describe terms: Bagging, Boosting and Stacking.	(Understand)	6
Q. 5 Solve Any Two of the following.		12
A) Explain clustering with example. State need of clustering.	(Understand)	6

- B) State the working of K-means clustering algorithm. (Remember) 6
- C) Differentiate between Agglomerative and Divisive hierarchical clustering. (Analyze) 6

*** End ***